

Table 1. SLC-StepDPO with SFT-anchor weight (λ_{sft})

test set: 360 solutions (60 problems $\times$ 6 personas), GPT-4o judge

Model	Final Acc.	Step Acc.	Explanation Match	Belief-Flip
SFT (Baseline)	73.9	91.5	79.5	8.3
Step-DPO	72.2	90.9	79.1	10.0
SLC-StepDPO ($\lambda_{sft} = 0$)	72.8	91.6	81.7	10.0
SLC-StepDPO ($\lambda_{sft} = 0.01$)	73.3	92.0	80.9	10.0
SLC-StepDPO ($\lambda_{sft} = 0.03$)	76.1	91.4	78.6	15.0

Lowering λ_{sft} trades Final Acc. for Explanation Match; best per column in bold, proposed shaded.